

0.1 The Forward Model

The problem is framed as an inverse problem. The forward map takes position and orientation data and calculated 8 flux values from each of the coils. Our goal is to approximate an inverse to this map. To do so, we first try to understand the forward map

0.1.1 Definition and properties.

First we define a pose vector $\mathbf{p} \in \mathbb{R}^5$

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \quad (1)$$

In actuality, $\mathbf{p}$ is restricted to a set $\Omega \subset \mathbb{R}^5$

$$\Omega = \left\{ \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \in \mathbb{R}^5 : \begin{array}{l} x \in (-0.125, 0.125) \\ y \in (-0.125, 0.125) \\ z \in (0, 0.25) \\ \theta \in (0, \pi) \\ \phi \in [0, 2\pi) \end{array} \right\} \quad (2)$$

Then we define a vector of fluxes, $\Phi \in \mathbb{R}^8$

$$\Phi = \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \\ \Phi_5 \\ \Phi_6 \\ \Phi_7 \\ \Phi_8 \end{bmatrix} \quad (3)$$

Now we can write the forward map: $\Phi(\mathbf{p}) : \Omega \subset \mathbb{R}^5 \rightarrow \mathbb{R}^8$. More precisely,

$$\Phi_i = \mathbf{H}_i(x, y, z) \cdot \hat{n}(\theta, \phi) \quad (4)$$

where $\mathbf{H}_i$ is the magnetic field calculated at (x, y, z) due to coil i using equation (??), and $\hat{n}(\theta, \phi)$ is the vector normal to the sensor.

Since this maps from a subset of $\mathbb{R}^5$ to $\mathbb{R}^8$ it is clear that the forward map is not surjective.

0.1.2 The Jacobian

Now that we have established that Φ is not surjective we must move our attention to thinking about local invertibility rather than global. To do so we define the Jacobian:

$$J(\mathbf{p}) = \frac{\partial \Phi}{\partial \mathbf{p}} \in \mathbb{R}^{8 \times 5}, J_{i,j} = \frac{\partial \Phi_i}{\partial p_j} \quad (5)$$

Since $\mathbf{H}$ is a finite sum of Biot Savart applications it is a smooth map away from the current filaments. $\hat{n}$ is also a smooth map, therefore Φ will also be smooth away from the current filaments and the Jacobian will be well defined there.

A closed form for $\mathbf{J}$ is cumbersome find so we instead use finite differences to approximate $\mathbf{J}$

$$\mathbf{J}_{:,i} \approx \frac{\Phi(\mathbf{p} + h * \hat{e}_i) - \Phi(\mathbf{p})}{h} \quad (6)$$

where $\hat{e}_i$ is the unit vector in the i th direction, and h is a small real value ($h = 10^{-5}$ in our simulations).